Task 1. Create GitHub repository and Node.js application

Create a public repository:

`insta-caption-bot`

GitHub: GitHub

Create these files:

```
None
insta-caption-bot/
│
├── app.js
├── package.json
└── Dockerfile
```

app.js

```
None
console.log("Hello, Instagram!");
```

package.json

```
None
{
  "name": "insta-caption-bot",
  "version": "1.0.0",
  "description": "Docker Node.js application",
  "main": "app.js",
  "scripts": {
    "start": "node app.js"
```

```
  }
}
```

Dockerfile

```
None
FROM node:22-alpine

WORKDIR /app

COPY package.json .

COPY app.js .

CMD ["npm", "start"]
```

Build the image:

```
None
docker build -t insta-caption-bot .
```

Run:

```
None
docker run --rm insta-caption-bot
```

Task 2. Create GitHub Actions workflow

Create this folder structure:

```
None
.github/
   workflows/
      docker-build.yml
```

Add the following YAML:

```
None
name: Docker Build Pipeline

on:
  push:
    branches:
      - main

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: Build Docker Image
        run: |
          docker build -t insta-caption-bot .
```

This workflow automatically builds the Docker image whenever you push changes to the main branch.

Task 3. Push Docker image to Docker Hub

Create a Docker Hub account:

Docker Hub

In your GitHub repository:

1. Open Settings.
2. Select Secrets and variables.
3. Select Actions.
4. Click New repository secret.

Create two secrets:

Secret name	Value
DOCKERHUB_USERNAME	Your Docker Hub username
DOCKERHUB_TOKEN	Your Docker Hub access token

Use a Docker Hub access token rather than your account password.

Update `docker-build.yml`:

```
None
name: Docker Build and Push

on:
  push:
    branches:
```

```
      - main

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: Login to Docker Hub
        uses: docker/login-action@v3
        with:
          username: ${{ secrets.DOCKERHUB_USERNAME }}
          password: ${{ secrets.DOCKERHUB_TOKEN }}

      - name: Build and Push Docker Image
        uses: docker/build-push-action@v6
        with:
          context: .
          push: true
          tags: ${{ secrets.DOCKERHUB_USERNAME
}}/insta-caption-bot:latest
```

Push the changes:

```
None
git add .
git commit -m "Add Docker CI CD pipeline"
```

```
git push origin main
```

Check the Actions tab in GitHub.

Task 4. Generate random Zomato-style food reviews

Update `app.js`:

```
None
const reviews = [
  "Amazing food and excellent service!",
  "Delicious pizza with great taste!",
  "Best biryani experience ever!"
];

const randomReview =
  reviews[Math.floor(Math.random() *
reviews.length)];

console.log("Zomato Food Review:");
console.log(randomReview);
```

Rebuild locally:

```
None
docker build -t insta-caption-bot .
```

Run:

```
None
docker run --rm insta-caption-bot
```

Example output:

```
None
Zomato Food Review:
Delicious pizza with great taste!
```

Push the changes:

```
None
git add .
git commit -m "Add random food reviews"
git push origin main
```

GitHub Actions will automatically build and push the updated image.

Task 5. ChatGPT prompt and final YAML

Prompt used:

Create a GitHub Actions CI/CD workflow for a Node.js application named insta-caption-bot. The workflow should automatically run whenever code is pushed to the main branch. Use GitHub Actions to check out the repository, log in to Docker Hub using GitHub Secrets, build the Docker image, and push it to Docker Hub. Use secure practices and explain each step.